版本号	修订日期	修订记录
V1.0.0.0	2021-11-18	初始版本
V2.0.1.1	2021-12-31	控制模块由CCU6变为GTM，更改过流触发条件
V2.0.2.1	2022-4-7	更换预驱，更改过流触发和使能条件
V2.0.2.2	2022-4-20	删除GTM输出引脚跳线
V2.0.2.3	2022-5-19	修改测压错误
V3.0.0.0	2022-12-13	重绘，支持6S输入
V3.0.1.0	2022-12-27	修改覆铜，修改霍尔滤波
V3.0.1.1	2023-12-14	修改结构框图错误

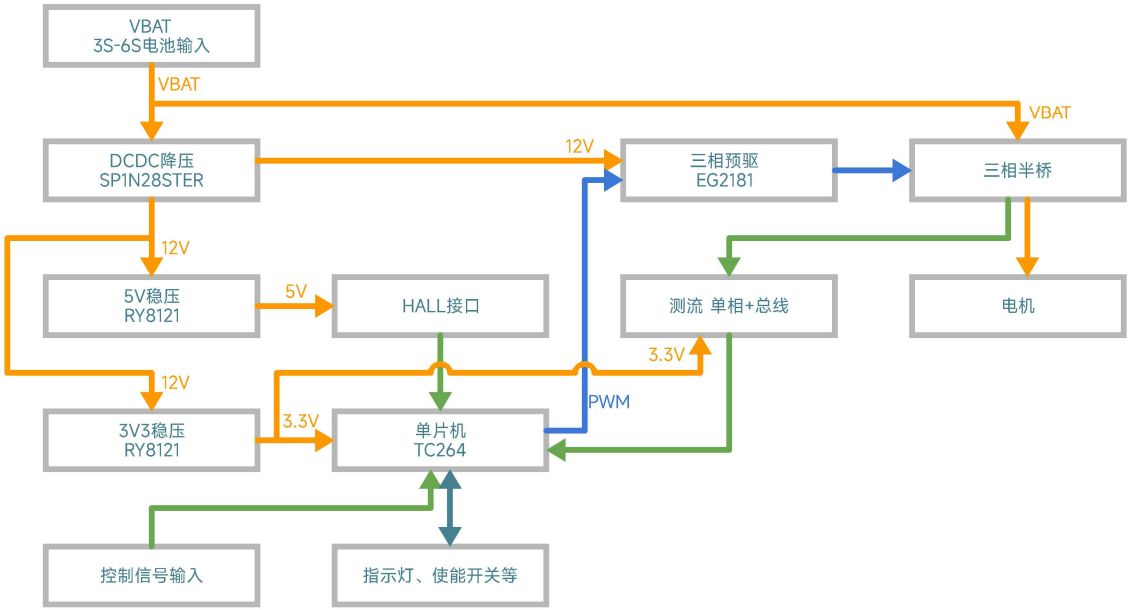
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本公司致力于产品的不断改善和功能升级，内容如有改动，恕不另行通知。
在此文件中，仅以版本号前两位(发行版)，划分版本之间的区别。

Title: 无刷驱动-Infineon-AURS2181S-GTM | **Author:** 猫肆管

Subtitle: Revision | **Revision:** V3.0.1.1

Size: A4 | **Date:** 2023/12/14 | **Time:** 11:43:48 | **Sheet:** 1 **of:** 5

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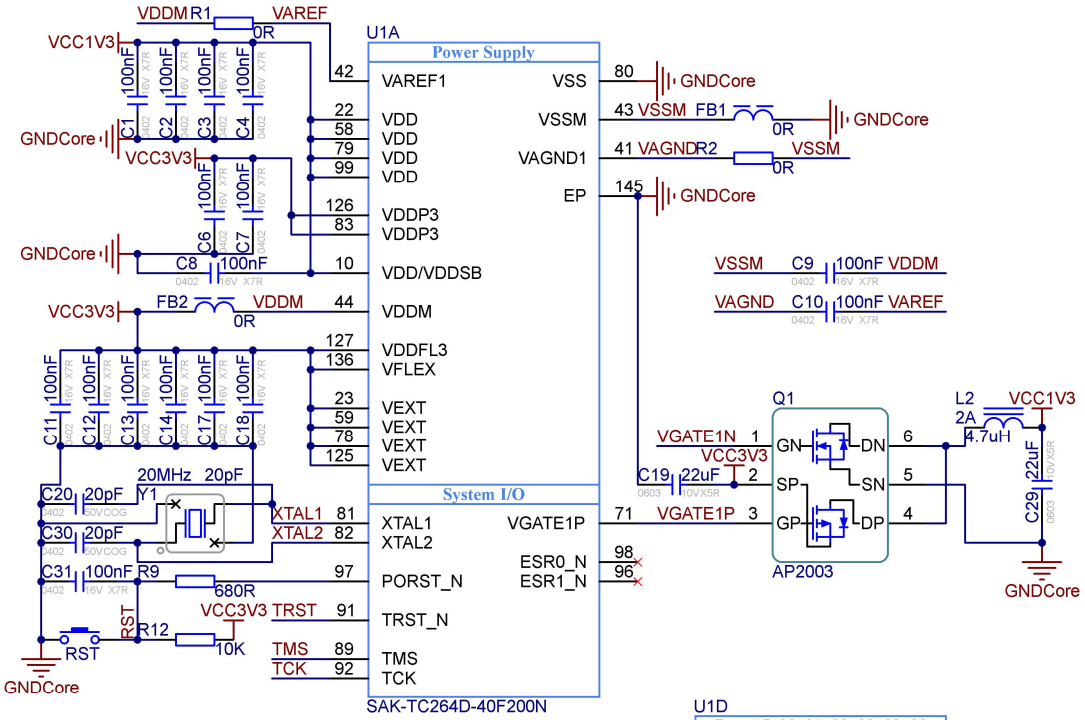
图例：

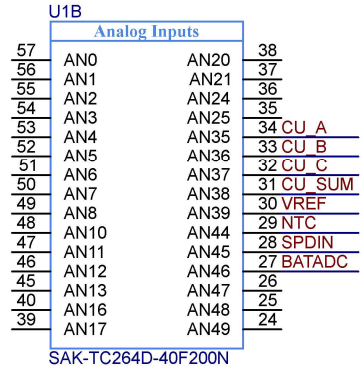
双向信号： UARTx　输出信号： UARTx　输入信号： UARTx　电源： 5V

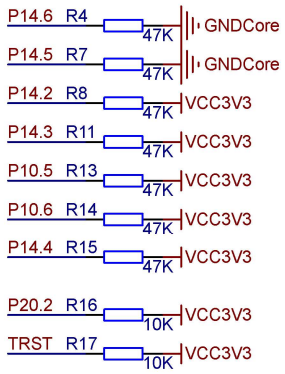
红色字体：需要注意/未确定条目

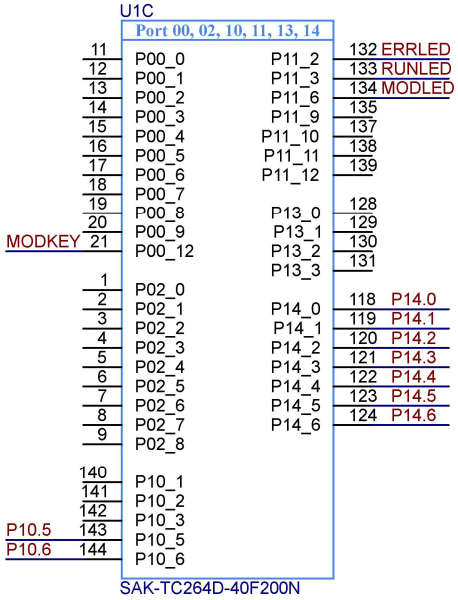
Title: 无刷驱动-Infineon-AUIRS2181S-GTM				**Author:** 猫肆管
Subtitle: BlockDiagram				**Revision:** V3.0.1.1
Size: A4	**Date:** 2023/12/14	**Time:** 11:43:48	**Sheet:** 2 **of:** 5	

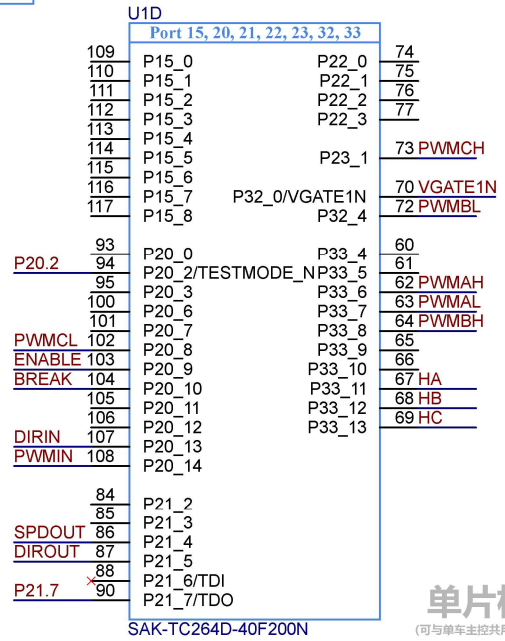
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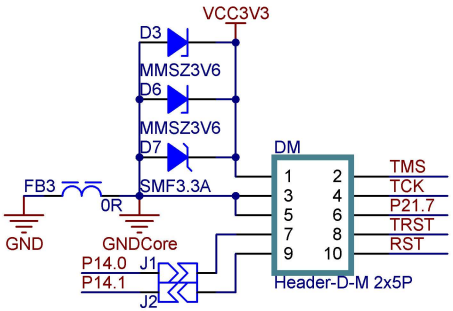




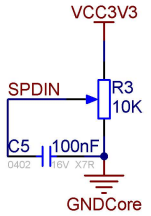




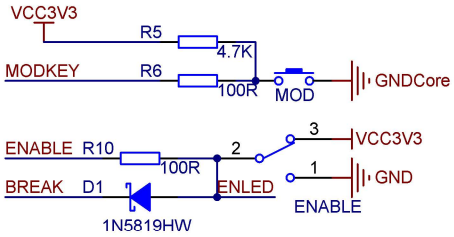




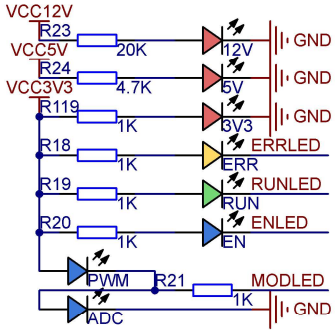
调速旋钮



模式切换按键&使能开关



信号指示灯



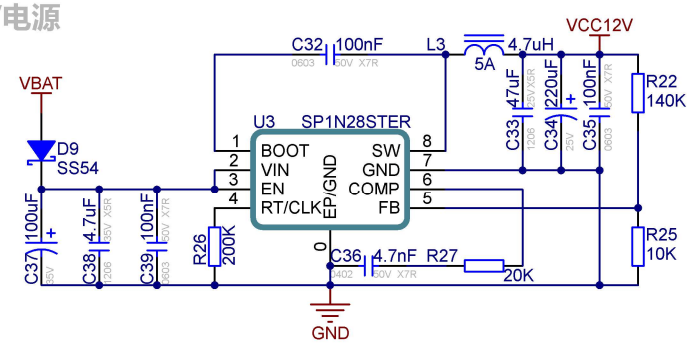
单片机与调试口

(可与单车主控共用,本板自带单片机仅为方便调试)

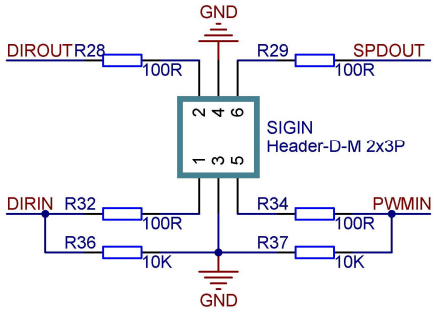
Title: 无刷驱动-Infineon-AURIS2181S-GTM			**Author:** 猫肆管
Subtitle: Control			**Revision:** V3.0.1.1
Size: A4	**Date:** 2023/12/14	**Time:** 11:43:48	**Sheet:** 3 **of:** 5

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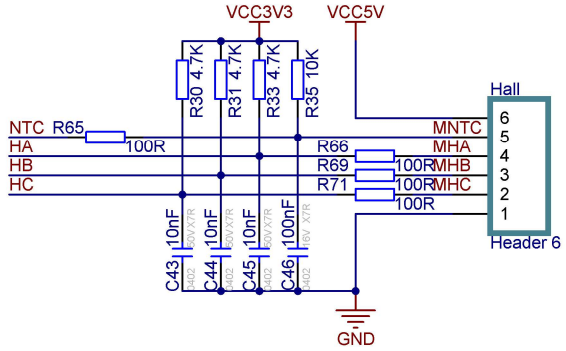
12V电源



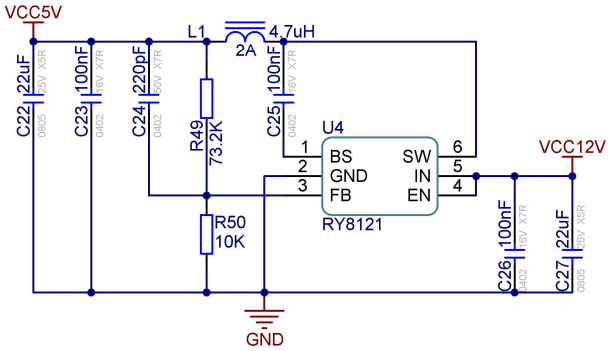
信号输入输出接口



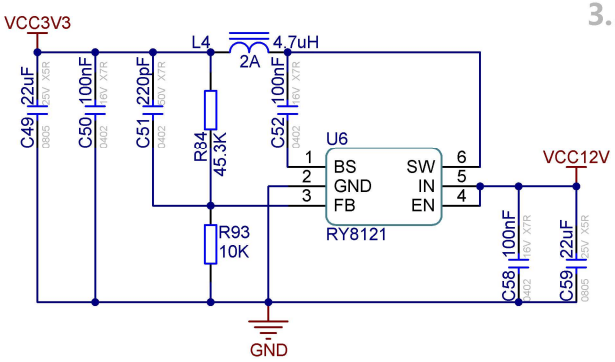
霍尔及温控接口



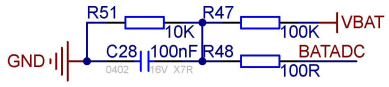
5V电源



3.3V电源

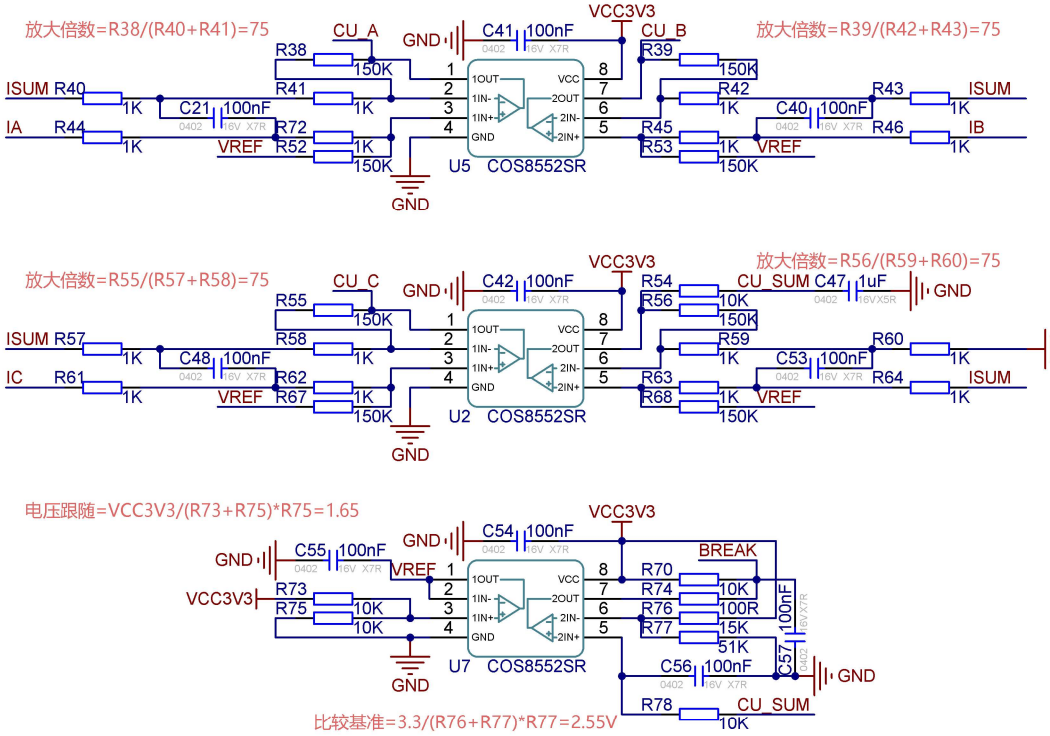


电源电压监测



电流放大

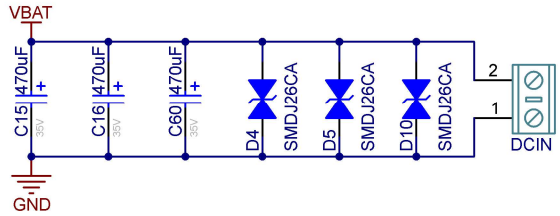
(采样放大倍数需按电机参数进行选择)



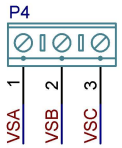
Title: 无刷驱动-Infineon-AUIRS2181S-GTM			**Author:** 猫肆管
Subtitle: Power			**Revision:** V3.0.1.1
Size: A4	**Date:** 2023/12/14	**Time:** 11:43:48	**Sheet:** 4 **of:** 5

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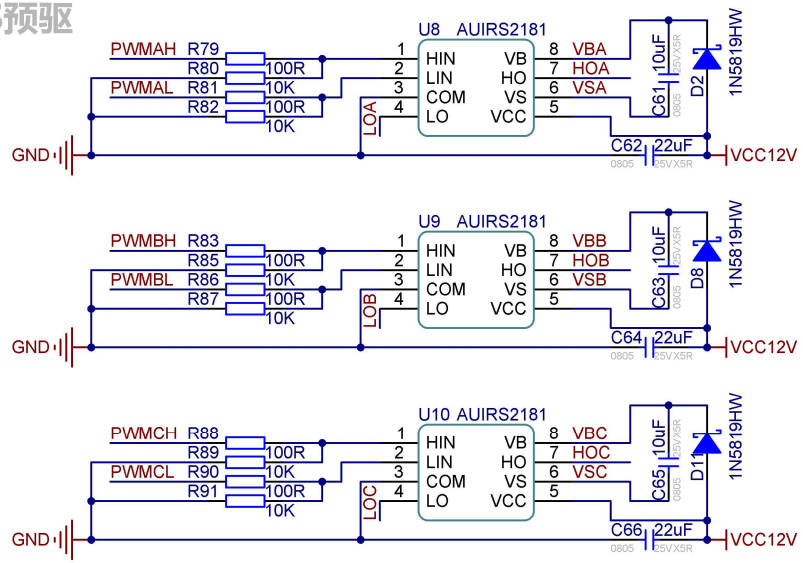
TVS及电容



电机接口

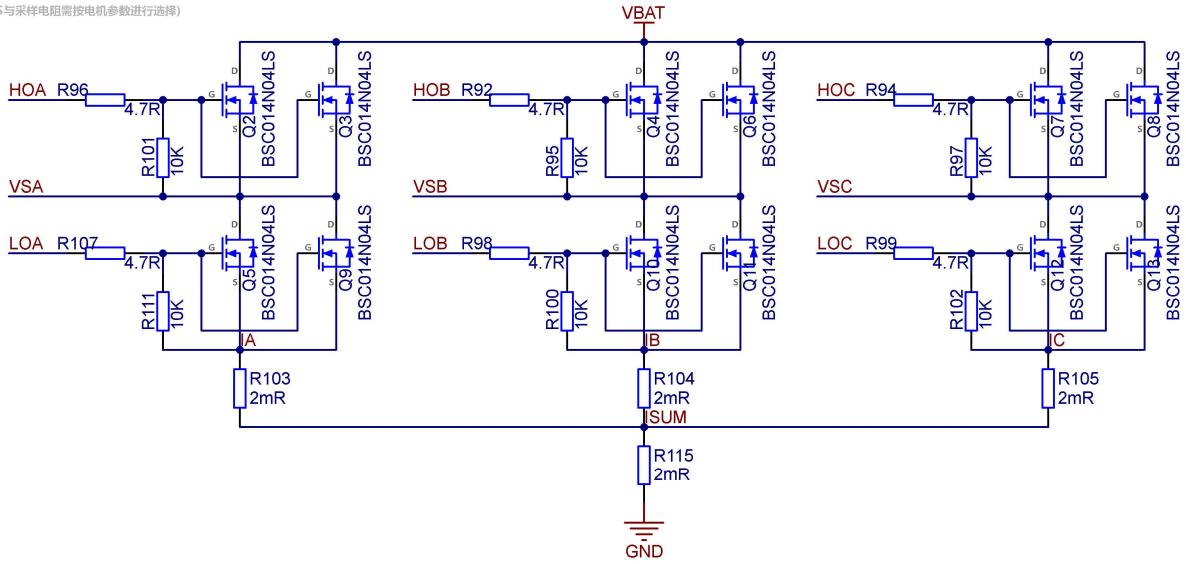


MOS预驱



MOSFET

(MOS与采样电阻需按电机参数进行选择)



特别说明:

该开源项目仅用于各参赛同学在设计自己的小车时进行参考，
硬件和软件均不能够直接拷贝使用在自己的作品中，
请大家参考学习后重新完成硬件和软件的设计制作。

Title: 无刷驱动-Infineon-AUIRS2181S-GTM			**Author:** 猫肆管
Subtitle: Output			**Revision:** V3.0.1.1
Size: A4	**Date:** 2023/12/14	**Time:** 11:43:49	**Sheet:** 5 **of:** 5

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